

A Packing of 79 Lines in $\mathbb{R}^{13}$

Limin Ge*

October 8, 2026

Abstract

We give 79 1-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 0.897009. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.896308, obtained by deleting 1 of the 80 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 79 subspaces of dimension 1 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.897009 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 1$. The cell $N = 79$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.896308, obtained by deleting 1 of the 80 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(13 - 1)/13 \cdot 79/78 = 0.934911$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N79.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 76 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
897009012308406090222533691510327214628841815339
denominator:
999999999998149516906048800670198724986766812335

B Coordinates

79 blocks of 13 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m13n1N79.txt`.

-191909786908	73833305224	-151516000622	148086843641
419211168953	5366795664	264346018728	270791675200
91387937527	-26659669207	9531626101	-275135850381
-124952386466	-229598979426	-160616566280	336646232410
290664527424	-311949875578	-29923578852	364496566745
84552734261	433070344640	52713494102	-4209997161
-214478114394	-80404077859	595335271727	422851077814
-435935393086	66441030906	44830925992	50599300505
193356307670	-243235042807	185309667601	52440340399
179521540445	-241141378495	-305766900618	-407662738815
-376511808757	666833546030	-49252267871	137602508574
13353242587	153995163375	-4984995237	415653052220
473537874104	76152762541	-523894886809	266935481721
363978985973	215340726775	318441616177	118541200212
-105934950206	-110663971856	315816777044	81356035058
51881875845	-119500650936	-90550700873	-160440397080
-127581031263	-166076170339	-332221234573	10806159490
299088376490	-254974537230	22084202543	-729257345509
-262877112796	-427314611544	-255335285342	168097689939
-352292699120	231426617085	-414241997368	248935614820
-331288250645	82818498281	-42695249201	-142479387022
-279383011804	169659081855	93381965469	67854364058
272052287856	679396458688	-45636328842	-228082673550
170311957230	111776177895	-67029401462	-102376403546
308371452815	-460291490460	66238872524	123353407469
410781841084	-183959919456	572941380333	-38374439367
-7905192039	218981940714	325791525305	228843361142
201997946655	30443502447	390624799720	545612742624
63384037188	266250299818	209410702626	-252683407664
274169512990	227222515001	-221750830958	-142937099122
-97905494014	82875020914	-124794290816	153981500462
-133822334608	198882020828	324888873391	-489198393787
87015592493	70134520543	-386814363079	47118811505
403481913738	169388478998	13965279030	175476230243
152593630702	38592803487	-349687319889	-141590899656
-326002922359	-69402541694	36300600677	302530185424
610672116731	200693486824	367764826344	375977279764
-340023614716	-247915347018	-374874813306	-123578173009
252953343879	46636196175	177296148105	563131850721
-336367795706	-566212591163	477823213583	-48500932959
-152726770079	-351977585112	-69257271407	210728384204
56861165898	-210551982409	126306087467	346690786536
264885963979	241343909363	-408733444954	207331477445
91176737727	-178058535018	88368984323	-390148785907
53515602795	-533580997191	223638713241	-1524305290
109075114010	-136729592490	235709445214	-358769611198
-156669347596	-87215883742	365288629316	49547999453
69637050936	602373745862	-529882666738	341291614406
278579384996	195160625026	106821730786	-234427435983
547092105177	90694590552	-223797209667	53348784603
-502932621900	-67633319527	-15885135715	-182727998710
-328339618487	-59751850234	-420062189640	-95497974650
30981934188	-175211991817	36967053191	-438187335950
86938257614	523775614508	791767727	376725687224
-88642646720	69100532485	256467171200	225561414658
-567584666778	-388522218355	132554137001	-156979846165
302709301059	124343030183	148412935930	94689303744
360677083150	198262784739	-628750404437	-128578223739
155519460897	245010290467	-46610968626	-128458237437
-116436699609	84405242054	515036898901	252204461934
-400057958234	-223446215203	291060617657	243179365736
-31359486076	54289545183	-171688210647	-188659665411
205352165330	-124964069108	129148673708	589664226589
-155446244259	-195513455773	-57306357189	-5688067145
417904852058	-62105609132	217018544038	8424396444
-647186536943	786559653750	-201727380065	-464071797426
-155547009580	-136550385590	230548452170	-164730649687
-368792777419	140807494613	150134086945	-115739969889
-164410738968	-48202913132	407271224131	-220845980273
-198159650460	-370171124630	-218471372648	-268565539151

-12595367585	93816445862	372515922212	-539223014203
411904898373	-356960257521	-3705094713	120432313050
-553137102670	-250908606525	194954753342	-31935414865
79865840022	-146407475407	-96016071002	213487568961
197502013734	233866367464	587418495758	92370611117
-319266625413	-251166477172	351852632459	201936450925
160985629262	-649254485419	-237940965839	277740357
-346408522410	42321798069	-99364661824	358388083111
-242066275545	333108817052	38984883316	-595666385001
-335075465001	-150366501647	145033805781	233535522559
-134828476829	-12320301126	-7298004137	-193947433853
290591975412	-539802817552	90568629892	-52711763949
307978838546	306061590925	85397703183	-435729936269
-196450912566	366560834267	-362570489369	337849166547
-265228379609	-9798853623	191081198278	-186663573211
-282000807712	-102082417297	-190396733664	14942958477
-269622049451	524844206925	570153246514	355648758177
-369075471489	-73087939837	-37562913165	276634094774
-297259938921	-223763432954	264910871106	160344583290
-102972831759	9522217311	-22407880228	-27448020980
125994905560	73267621594	335291110487	520802784348
-181702631322	-126743516672	37005754729	-157001333174
265603360345	191250156374	-518917970476	327641394984
-120342648539	-232638492187	647635558904	113195327877
206215769995	296078166090	-286272376858	124127016144
-431345149554	-43482555269	-405774893185	146126339530
101406144033	-322818204497	37984585838	-8712634916
-373629804204	-467208568125	151464319370	-120256824347
-68222720148	-663317206327	-276235587948	1303302600
-340538997794	5604321318	157380388465	-499002362051
-80543349740	29474446807	-131140890567	-402850258993
-591526953869	78321546703	90141292623	88830203706
37279371322	-21306012996	-220202839274	514675176870
266233526235	-183608986115	-138346952979	67594517226
350369294548	145561096175	-141710148053	239977138547
-204275766642	394392938043	308981824590	-8115188937
508382993302	436044476185	190534749363	89148105243
220705500721	-122872836111	294749213433	458108570607
157117649879	-149259437256	459210118657	147966903952
-50717159296	-287513108506	450170643220	-97446053195
-24014239220	-121588786955	-419428910247	-104135997005
144629012711	57396913847	276072851612	341357768739
42764846793	-306995208014	5973208493	682275748841
635836421723	-189267743220	410195318802	-214441502754
-34775453869	-144084794222	-78338562304	-274181874797
123320449574	-322782153363	-33111719390	-50397737308
184353124057	489894039369	-42430344118	472760881492
-154498490453	-232598369681	163128919734	-65304909908
204876150259	561139583572	84938010194	52295276571
290468147303	-467917760747	-205009554150	-55685309422
-56896102440	-179688492639	-98392987212	-135780742313
162088258232	-26730322998	-136201058197	-128825047257
-345050410796	220147086069	14687631737	-201258612426
-647475204572	318064179183	479774628770	76111010674
-124337571848	-239355423162	-27820729589	275182022710
-411658457752	-210693485990	216383026861	193407775832
8622679246	-20421885570	204814920313	-158961831515
218631339491	-75412855274	67693884301	273107073875
548507061073	78550057717	-412449633075	134831967439
-17208896692	-339829783650	-509102254187	241569992540
-197785065128	129696506797	288411701615	66920514486
694886251025	315585252682	-304518337788	-623224743035
19890672382	300327535186	-224050439916	-298129571476
63160920443	-138659015189	-55499104969	407202337182
-131710944363	-4692723709	4457134626	-239378897147
-72593407418	148192355492	-245415757899	341660891875
-187164495414	652640542301	-391379317976	-413400869716
186327687682	-239515423014	300583513319	284458961511
203486382301	121571727988	153373283785	-565293598426
-110238189407	-272968014718	-638837618617	251407739451
-162934912972	-418635578426	342951318343	-297455119865
440577441781	14523758187	-146910182966	77781725337
326183994747	64137299209	151833329557	234071193488
-116218838021	-504276127983	193064884688	86117633370
-73200930919	37689575291	-108543931259	149778439746
-641837111649	-211077681406	-14988265246	110527871782
-313946576915	141211333061	-287037812693	766706436
84945280245	-407810152823	353835739601	-396719000017
-305085916974	-180759430996	7238535798	-326577247268
172818705727	24019917117	-193880733320	407422778582
-76254153567	-239281943780	495110090621	-44144567627
-154465172703	-100416273289	-279972618014	222404383183
-17482446594	-288325571881	227973502638	-251170647090
-99868662978	-533289030844	-279974495820	12419802141
-34482267132	204601177930	-137552932639	-332928166028
1384529277	59168074504	167185372650	392796391543
-380916882924	-117400794336	-499151746367	341853256346
446672777990	274050343250	-68134068745	25695623382
-154109398116	293689959584	-518404373471	231800994077
-191140624018	-51726562715	-112611355551	-137582100569
520338788944	142616324472	-264333099348	92880795727
-323019404875	159794607623	102360535508	-649691071948
115395421297	311060387975	198561964463	310543610649
-339789113547	214453034581	-479061268797	-226085090433
255371203657	-502941041794	179155030724	-43453676252
-55588932459	-523357084436	476301369641	324236765666
-147441690457	208949480757	263226316860	94269786687
307061964259	60645533379	148310918430	-124393835348
-99657349351	-243662750510	94319448421	-190653229518
-32390417280	109507910978	-50467914788	120847157995
-27300969122	-307383363618	-87249398804	332723792612
363694725859	-386795261239	-143644043256	358634109709

8264932113	-306268929629	195117687220	-368641312678
78525492749	-145487115458	-149173311383	304791299125
-153927098721	365181934373	17585661472	-74605874927
-236230673280	500087650527	37695948889	360729392397
-108326232216	-11394828256	66617035980	-30758183442
314522025259	-107048106488	-251096132246	-184549577985
31619310612	-477793708741	-81202018324	63719514604
93168819350	-89974642748	-333246315582	-53876894224
7083004118	127572930709	-512679075366	249144936067
-657969669502	348117261151	-345282973946	302295756826
418785619660	33310012886	463942291243	-275929055863
159937239926	-463970623102	-394106642127	299400300267
-63692086459	216314147520	13767284584	419937418400
-394973574874	-156328246620	-57404121026	-279150204857
-457737383933	-156137182280	179509718039	-235304875182
434092585762	122819613116	-368937020553	198158458580
-73140087449	294286337400	-545634260218	-61487951012
84837050478	-542701460018	-308212917390	119506388120
84589909444	-373560523219	100687869841	-553925825484
96160509804	-81824150971	-517627884469	-478431026444
185849255345	-380302915603	-232223830428	-495338030092
410895692029	436421066508	-133940787165	93089407187
-272021894165	66888610175	-40508183772	337198195964
-17346141123	92623333068	-214421203162	-437887699769
-182033893550	-345669113594	-188902812055	192868820116
165628715906	-174029477061	49073424541	118478570878
484613036860	488838876077	-574129828057	167726032471
-395878955442	-119558378051	-527172619	81531605135
231099123488	235094657813	125931314930	308484065411
-88144568605	384511138142	-95447132993	-13340439291
544562585983	84538572299	135389934189	-1003677982
391282793671	-145061251926	-113351199034	173993444537
34306555436	132341524408	79913985429	145907169286
101385980409	-352534667126	28849556321	-340530578991
-134935442120	-168516553848	-430567525264	-575934698630
-31628423294	-185467651960	-13006481557	53225391952
433687411003	-258134733846	-289099466008	-425346749416
-255229260850	-453021895565	-567401466597	35242141604
69398195895	-317732150097	-126205597580	-23328051619
-209155695573	27258402082	40245302158	-375965836094
161970101454	-110136274375	-253481964637	-318048649871
463408965054	-63845471210	-46725285057	127156018956
-168330045765	325151716340	373066194509	89680789492
251990912539	-11599649173	36404709487	266162516759
217449852950	-42028845138	8473171964	89169520746
66867342648	-561723467385	-124263748863	111698661096
18134158884	-78125917985	288850385701	-14968083009
493975624417	-158709387302	-318244654183	555942231602
153894162610	242273564855	219790323546	219069996813
-27266560539	-151900395170	-146814051460	274920684604
170488277367	-561123179992	719637978060	-253815543919
284998557923	-195392826059	-58922134871	-372425912267
-486178607499	-512320260617	-372752644832	8915191040
-226414776158	-130264706766	165272942358	-261631602118
-431570897559	52685685750	87133240171	82237737957
-209339606241	82474362215	398800659178	185484085690
35060321476	-2282430097	-111295795183	123199455032
205438896383	487483657691	481750784062	-476083071288
113478087648	85062390429	172397334998	-321542889270
-110742255985	39609028504	-356048166890	-52549781037
-241789972425	-104674752706	-87150704203	213867087394
236279305593	303099921455	-222535042775	9658040033
-188945762888	348329506884	-25444777532	-230860298391
-109587487813	117601763931	-336353387631	-254019068536
516335083884	601073606970	308620015847	-350728907747
470618964966	-354411630663	203149678714	591622862931
-191952248638	-149236624926	-24507691674	-97739058962
140271111547	286378825694	-505792744441	-478043569934
431374813892	387348513290	258131762409	-126959889636
-43210491176	69317519842	-154661316614	7647422276
-50729774487	119163220301	-131566469429	-53943153353
-484449660784	14696565170	416862980964	-12290796613
307075955779	-189783956728	260059518989	-464083609946
247633193290	151456869448	-122528208964	213501055254
-185331107364	-131647927405	227746216552	4471754902
214405261391	26842681712	-465002559715	169837679587
-415333207998	29792201107	-4669126286	33366650042
265194830194	291324096989	-264395602302	15865899633
-198738373030	-108312619778	239145669673	368964005314
-8522021575	458862132047	533406953266	131233290457
359941329028	235685200671	383711192707	377338223430
-37372250915	443548365175	315243739988	-286341424860
-275146446041	24342812327	21584820588	-302825022825
229080969618	-546363024046	61504766704	-489108975542
50006434677	252111052771	89074339378	